

Portfolio Optimization using Linear Optimization and Monte Carlo Simulation:

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Project Objective:

The objective of this project is to construct an optimal wealth management portfolio using historical market data and mathematical optimization techniques.

Historical monthly return data were collected for eight asset classes representing U.S. equities, international equities, real estate, gold, bonds, and cash equivalents over the period 2010–2025. The expected monthly return for each asset class was estimated using its historical average monthly return, while risk was measured using the historical standard deviation of monthly returns.

A client has \$1,000,000 available for investment. The wealth manager must determine which asset classes should be included in the portfolio in order to maximize expected monthly return while satisfying a set of investment policy constraints.

To ensure an appropriate balance between growth potential, income generation, and capital preservation, the asset classes are categorized as Growth, Income, and Defensive investments. The portfolio selection process is subject to the following constraints:

- Exactly 5 asset classes must be selected.
- No more than 2 Growth assets may be selected.
- At least 1 Income asset must be selected.
- The number of Defensive assets selected must be greater than or equal to the number of Growth assets selected.

Excel Solver is used to identify the portfolio that maximizes expected monthly return while satisfying all investment constraints. This solution is referred to as the Optimal Portfolio.

To evaluate whether alternative portfolio combinations could provide comparable performance, a second optimization model is developed. An additional diversification constraint is introduced requiring that no more than 3 of the 5 asset classes selected in the Optimal Portfolio (Optimal 1) may also appear in the Second-Best Portfolio (Optimal 2). This forces the model to identify a different portfolio composition while still maximizing expected return under the original investment policy constraints.

Monte Carlo simulation is then performed using 1,000 randomly generated scenarios based on the historical return and risk characteristics of each portfolio constructed (Optimal 1 and Optimal 2). The purpose of the simulation is to test the robustness of the optimization results

under uncertainty and determine whether the portfolio identified by the linear optimization model (Optimal 1) consistently outperforms the alternative portfolio (Optimal 1) across a wide range of possible market conditions.

The results provide insight into portfolio construction, diversification benefits, risk-return trade-offs, and the reliability of optimization-based investment decisions under uncertain market environments.

Results:

Optimal 1:

Model	
SPY (S&P 500 ETF)	1
QQQ (NASDAQ 100 ETF)	1
IWM (Russell 2000 ETF)	0
VEA (Developed International Markets ETF)	0
VNQ (Real Estate ETF)	1
GLD (Gold ETF)	1
BND (Total Bond Market ETF)	1
Treasury Bills (TB3MS)	0
Expected Return	4.68%

Based on the objective and constraints provided, the first optimization model selected assets were **SPY, QQQ, VNQ, GLD, and BND** and the expected optimal expected return for the month came out as **4.68%**.

Optimal 2:

Model	
SPY (S&P 500 ETF)	0
QQQ (NASDAQ 100 ETF)	1
IWM (Russell 2000 ETF)	1
VEA (Developed International Markets ETF)	0
VNQ (Real Estate ETF)	1
GLD (Gold ETF)	1
BND (Total Bond Market ETF)	0
Treasury Bills (TB3MS)	1
Expected Return	4.45%

For the 2nd best optimal asset mix, in addition to the existing constraints in optimal 1, we added an additional constraint that no more than 3 of the 5 asset classes selected in the Optimal Portfolio (Optimal 1) may also appear in the Second-Best Portfolio (Optimal 2).

This gave us the asset mix of **QQQ, IWM, VNQ, GLD, and TB3MS** as the optimal portfolio mix, with the monthly expected return of **4.45%**.

Simulations conclusion (1000 uncertainty test):

After testing the expected returns of both optimal solutions in 1000 randomly generated scenarios, we got the following results:

	Optimal	2nd Best		
Mean	4.23%	4.15%		
SD	9.59%	10.35%		
Probability (Optimal > 2nd Best)	=	51.20%		

The average of all 1000 expected returns for optimal 1 is slightly better than optimal 2, while Standard deviation is not much different. When we checked the probability of optimal solution 1 having monthly return greater than optimal solution 2, optimal solution 1 is better with a very thin margin.

Conclusion:

Portfolio 1 is recommended because it achieved the higher expected return and marginally outperformed Portfolio 2 in Monte Carlo simulations. However, the small difference suggests that both portfolios are viable investment choices and perform almost similarly under uncertainty.

Note: *The Monte Carlo simulation results may change slightly each time the workbook is recalculated because the model uses Excel's RAND() function to generate random scenarios. This is expected and desirable, as it enables the simulation to assess portfolio performance across many different possible market conditions.*